

Simon Gardier

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About Me

M.Sc. in Computer Science (Intelligent Systems, Cum Laude) from the University of Liege, Belgium. Second author of a paper accepted at **CVPR 2026** (Toyota Motor Europe). Experience in computer vision, vision-language models, embedded systems, and robotics.

Areas of interest: Embedded systems, Robotics, Machine learning, Deep learning, Computer vision, Cloud computing, Parallel programming.

Education

University of Liege

Liege, BE

M.Sc. in Computer Science (Intelligent Systems)

Sep 2024 - Jun 2026

- Relevant coursework: Machine Learning, Deep Learning, Theory of Computation, Intelligent Robotics, Compilers, Information Theory, Computer Vision, Software Engineering Project, Machine Learning Systems Design, Distributed Systems, Mobile Application Development, and more.
- Graduated Cum Laude.

University of Liege

Liege, BE

B.Sc. in Computer Science

Sep 2020 - Sep 2024

- Relevant coursework: OS, Discrete Mathematics I&II, Statistics I&II, Logic, Computation Structures, Parallel Programming, Introduction to AI, Embedded Systems, DSA, OOP, Computer Networking, and more.

Experience

EVS Broadcast Equipment

Dec 2025 - Jun 2026

Master's Thesis — R&D in Computer Vision

- Designed and trained a modular feed-forward 3D reconstruction pipeline for soccer broadcast scenes.
- Achieved novel-view synthesis at ~1s per scene on a single GPU.
- Benchmarked 3D foundation models on a custom evaluation pipeline.
- Reviewed the state of the art in feed-forward novel-view synthesis and 3D reconstruction.

Toyota Motor Europe

Sep 2025 - Dec 2025

AI Research Intern (Paper accepted at **CVPR 2026**)

- Implemented a proof-of-concept for a new VLM personalization method presented at Toyota's Innovation expo.
- Evaluated and performed ablations (100+ experiments) for the proposed method.
- Implemented the method and its testing suite in Python using PyTorch.
- Collaborated on the writing (text, tables, figures) and submission of the scientific paper to CVPR (2nd author).

RoboCup Montefiore Team

Sep 2023 - Jun 2026

Software Engineer

- Led the computer science team in the RoboCup project at Montefiore Institute, University of Liege.
- Developed a bootloader for the ARM Cortex M4 with libopenm3.
- Built a sim-to-real pipeline to control the robot from a virtual clone in Blender.

University of Liege

Sep 2021 - Sep 2024

Teaching Assistant

- Assisted a class of ~20 students during the practical sessions of the INFO2056 Python course.

Projects

Personal website & blog

2026

- Designed and developed a portfolio and blog with Astro, SolidJS, and UnoCSS.
- Implemented interactive components (search with Fuse.js, animated hero) as SolidJS islands, RSS feed and sitemap generation.
- Hosted on Google Cloud.
- Tools: Astro, SolidJS, TypeScript, UnoCSS, Google Cloud.

Monte Maps (Indoor Google Maps without GPS)

2026

- Implemented a Google Maps-like application localizing the user in the Montefiore Institute (ULiege) from an off-the-shelf camera, on-device and in real time.
- Built the localization pipeline from image retrieval (MegaLoc, FAISS) and a COLMAP point cloud.
- Released the dataset (point cloud, mapping images, descriptors) on HuggingFace.
- Tools: Python, PyTorch, COLMAP, MegaLoc, FAISS, Viser.

MineTexture: Diffusion for Minecraft texture generation

2026

- Built an end-to-end ML platform generating Minecraft-style textures from user prompts.
- Covered the complete MLOps lifecycle: dataset creation and preprocessing, model training (LoRA fine-tuning of Stable Diffusion 1.5), cloud storage, API serving, Docker deployment and a web interface.
- Tools: Python, PyTorch, FastAPI, Stable Diffusion 1.5, LoRA, Docker.

Bootloader for Cortex M4 MCU with updates over CAN bus

2025

- Designed and implemented a bootloader for the STM32F303 MCU of the servomotors of the RoboCup Montefiore Team robots, with firmware updates over CAN bus.
- Documented the STM32F303 MCU architecture and the bootloader design.
- Tools: C, libopencm3, arm-none-eabi, openocd.

YOLOv5 (m) from scratch

Mar 2025 - May 2025

- Reviewed technical reports from Ultralytics to implement the model with PyTorch.
- Implemented the Backbone and the Neck of the model.
- Trained the model on a labeled dataset of 3k Minecraft gameplay screenshots for ore detection, on Google Colab with Weights & Biases.
- Tools: Python, Scikit-learn, PyTorch. Grade: 15/20.

Dance Dance Revolution in assembly

Mar 2024 - Jun 2024

- Implemented embedded software (~7000 lines of assembly) for PAL analog video signal generation and a music player (104x144 images at 50Hz on a 32MHz CPU with only 2048 bytes of RAM).
- Developed a Dance Dance Revolution game on top of the embedded software, in Assembly, with a custom-built input pad.
- Tools: MPLAB X IDE, PIC Assembly, PWM, PAL standard. Grade: 20/20.

Other projects

Delivery robot with ROS2, Risk Android Game, Theoretical study of Machine Learning classifiers, Flashcard TUI, physics simulations in Python, Ford-Fulkerson visualizer with NetworkX, Harvard computer circuit design, Mini C Standard Library from scratch, DTW for drawing classification,...

Skills

Programming Languages:  Python,  C,  Java,  C#, SQL,  Dart, TypeScript, ARM/PIC assembly.

Tools and Frameworks: PyTorch, Scikit-learn, OpenCV, Docker, OpenShift, Google Cloud, Git, Jira, Linux (Arch, Fedora, Ubuntu, Raspberry Pi OS), OpenMP(I), Flutter, $\text{\LaTeX}$

Languages: French (native), English (proficient), Dutch (beginner)